Linear algebra – 10

Matrix norms and localisation of the spectrum

Homework

12 сентября 2026 года

Theoretical minimum. Operator and Frobenius norms; the spectral radius and Gelfand's formula; Gershgorin discs; Courant–Fischer and Cauchy interlacing; Weyl's inequalities $\lambda_{i+j-1}(A+B) \leqslant \lambda_i(A) + \lambda_j(B)$; the Hoffman–Wielandt inequality; the numerical range $W(A) = \{\langle Ax, x\rangle : \|x\| = 1\}$ and the condition number $\varkappa(A) = \|A\|\,\|A^{-1}\|$. The roots of a polynomial depend continuously on its coefficients, so the spectrum and the spectral radius $\rho(A)$ are continuous functions of the entries of A. A continuous function f with $f\left(\frac{x+y}{2}\right) \leqslant \frac{f(x)+f(y)}{2}$ for all x, y is convex.

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1. Prove Gershgorin's theorem together with the statement about connected components, and deduce that a strictly diagonally dominant matrix is invertible. Give an example showing that the union of the discs cannot be replaced by their intersection with the corresponding column discs.

2. Prove the Courant–Fischer minimax theorem, deduce Cauchy's interlacing theorem, and derive Weyl's inequality $\lambda_{i+j-1}(A+B) \leqslant \lambda_i(A) + \lambda_j(B)$ for Hermitian matrices.

3. (VJIMC 2003, cat. I) Let A and B be complex Hermitian 2×2 matrices with eigenvalue pairs (α_1, α_2) and (β_1, β_2). Determine all possible eigenvalue pairs (γ_1, γ_2) of $C = A + B$.

4. (VJIMC 2012, cat. II) Let M be the tridiagonal 10×10 matrix with -1 in the corner m_{11}, with 3 in the positions m_{12} and m_{21}, with 2 on the rest of the main diagonal and -1 on the rest of the two neighbouring diagonals. Show that M has exactly nine positive real eigenvalues, counted with multiplicity.

5. (SEEMOUS 2012) Let k be a positive integer and let A be a real symmetric $n \times n$ matrix with $\big(\operatorname{tr}(A^k)\big)^{k+1} = \big(\operatorname{tr}(A^{k+1})\big)^k$.

   a) Prove that $A^n = \operatorname{tr}(A)\, A^{n-1}$ when k is even.

   b) Does the same conclusion hold for odd k?

6. (SEEMOUS 2018) Let $A, B \in M_{2018}(\mathbb{R})$ satisfy $AB = BA$ and $A^{2018} = B^{2018} = E$. Prove that if $\operatorname{tr}(AB) = 2018$, then $\operatorname{tr} A = \operatorname{tr} B$.

7. (after IMC 2025 shortlist) Let n be a positive integer. Find the least positive integer k with the following property: for every complex $n \times n$ matrix A at least one of the k matrices $E + A,\ E + A^2,\ \ldots,\ E + A^k$ is invertible.

8. (IMC) Let A and B be invertible complex $n \times n$ matrices with $A + B = E$ and $(A^2 + B^2)(A^4 + B^4) = A^5 + B^5$. Find all possible values of $\det(AB)$.

9. (Romania national olympiad 2007) For $n \geqslant 2$ let H_n be the set of column vectors $x \in \mathbb{R}^n$ with $\sum_{i=1}^n |x_i| = 1$. Prove that there are only finitely many matrices $A \in M_n(\mathbb{R})$ for which the map $x \mapsto Ax$ satisfies $A(H_n) = H_n$.

10. Prove the Hoffman–Wielandt inequality $\sum_i |\lambda_i(A) - \lambda_{\pi(i)}(B)|^2 \leqslant \|A - B\|_F^2$ for normal A, B and a suitable permutation π, and deduce the standard bound $\|\delta x\|/\|x\| \leqslant \varkappa(A)\, \|\delta b\|/\|b\|$ for perturbations of the system $Ax = b$.